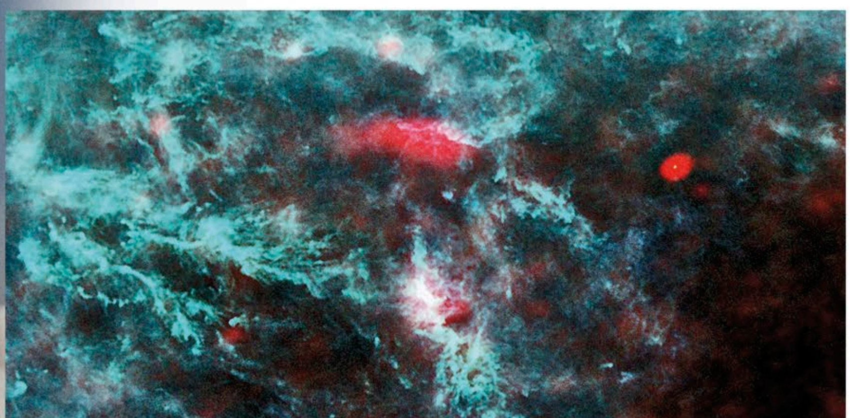
INFRARED AND MICROWAVE

Some infrared and microwave radiation penetrates Earth's atmosphere but to detect the full range requires observing from space. Prominent targets for observation are cool stars and active galaxies, which emit much of their radiation in the infrared. Infrared telescopes also make it possible to see through interstellar dust clouds into regions obscured from optical view, such as the interiors of nebulae and the center of our galaxy. The largest infrared telescope launched so far was the Herschel Space Observatory, which had a mirror 11.5 ft (3.5 m) across and operated from 2009 to 2013. Microwave space telescopes are designed primarily to detect and map the cosmic microwave background radiation, in order to investigate the structure and origin of the universe. The first dedicated microwave space telescopes were the Cosmic Background Explorer, launched in 1989, and the Wilkinson Microwave Anisotropy Probe, launched in 2001. The most recent was the Planck space telescope, which was launched in 2009.

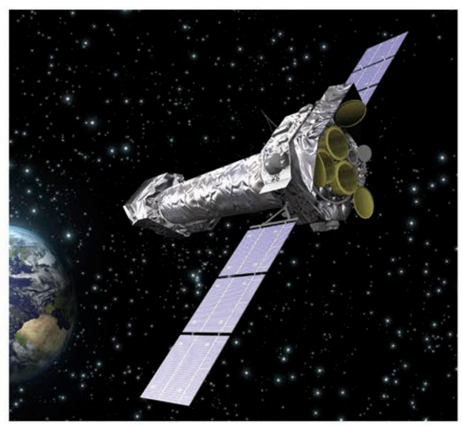
PLANCK IMAGE OF STAR-FORMATION REGION IN PERSEUS
This false-color image of a low-activity star-formation region was produced by combining data from Planck at three different microwave wavelengths.



PLANCK SPACE TELESCOPE
Shown here being tested before its launch in 2009, the Planck satellite carried a telescope with a 4.9 ft (1.5 m) main mirror to study the cosmic microwave background radiation. During a mission that lasted until 2013, it measured the age of the universe as 13.8 billion years.

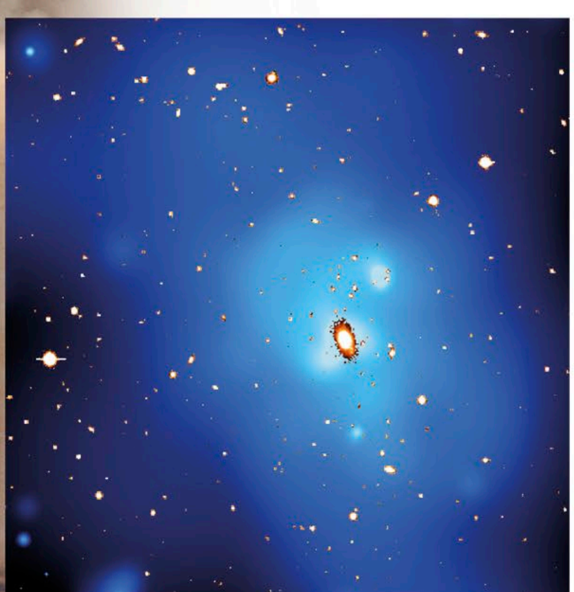
X-RAYS AND GAMMA RAYS

The shortest wavelengths of all, X-rays and gamma rays, are produced by some of the most violent events in the universe, such as supernova explosions. However, like infrared and microwave radiation, X-rays and gamma rays are best studied from space. Major X-ray space observatories include the Chandra X-ray Observatory (see p.37) and XMM-Newton, both launched in 1999, and the NuSTAR, launched in 2012. Notable gamma-ray space telescopes include the Compton Gamma Ray Observatory (see p.37), launched in 1991, and the Fermi Gamma-ray Space Telescope, which was launched in 2008 and carries an instrument designed to study gamma-ray bursts, which are thought to be emitted by the merger of black holes and neutron stars and also by the collapse of massive stars to form black holes.



XMM-NEWTON X-RAY SPACE TELESCOPE
Launched in 1999, the XMM-Newton contains three X-ray telescopes for the imaging and spectroscopy of X-ray sources. The entire satellite is 33 ft (10 m) long and is in a highly elliptical orbit that, at its most distant, takes the satellite more than 60,000 miles (100,000 km) from Earth.

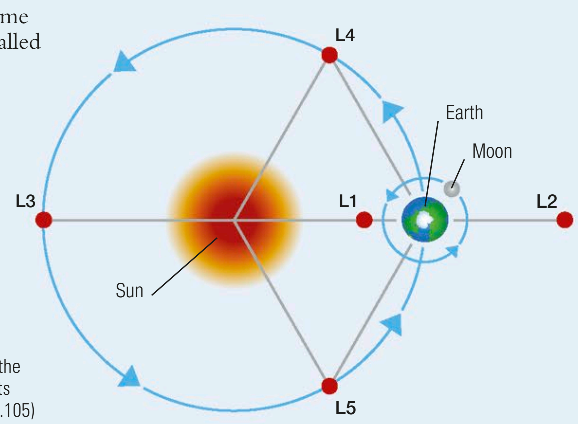
X-RAY-EMITTING CLOUD
This XMM-Newton image shows an X-ray-emitting cloud of ultra-hot gas, at temperatures up to about 90 million °F (50 million °C), around a giant elliptical galaxy.



EXPLORING SPACE

LAGRANGIAN POINTS

Satellites can be placed in various orbits around the Earth or other celestial objects. Some satellites are placed at specific points called Lagrangian points. These are locations where the orbital motion of a small object (such as a satellite) and the gravitational forces acting on it from larger bodies (such as nearby planets and stars) balance each other. As a result, the small object remains in a fixed position relative to the larger bodies. There are five such points in the Earth–Moon–Sun system.



FIXED ORBITS
This diagram shows the five Lagrangian points in the Earth–Moon–Sun system. Satellites at these points orbit the Sun, not Earth, and include SOHO (see p.105) at L1, and Herschel, Planck, and Gaia at L2.